Rules and Regulations



Academics

Become a Student

Life at Trine

Student Support

More in this Page

Competition Information

Rules and Regulations

Event Agenda

Event Location

Team Registration

Technical Design Report Submission

Participating School Funding Documents

Honored Guest Registration
TrineOnline →

Event Volunteer Registration

STEM Activity Day

Team Media Submission

Student Resume Submission

Sponsorship Opportunities

Competition Winners

1. Team must submit Participant Roster by September 25, 2026.
2. The team must submit Technical Design Report by April 2, 2027, at midnight.
3. The team must pass the safety inspection and check in before putting LPV in water.

About Trine

News

Give

Athletics

Alumni

International

Academics

Become a Student

Life at Trine

Student Support

4. The teams will comply with all rules provided in the competition.
5. Point bonuses will be awarded for full autonomous LPVs during the competition. Partial points will be provided if remote or human intervention is required during the competition.
6. The team must be able to remotely pilot boat in emergency situations to return boat to shore.
7. **Competition Vehicle Design**

    1. The team may integrate all or portions of the given sensor suite to the LPV along with any additions they deem necessary to complete the task.
    2. Teams MAY use the provided LPV hull as a **base platform**.
    3. The team's competition LPV hull may be:

        1. Used "**as-is"** from the originally provided hull,
        2. Teams MAY use the provided LPV hull as a **base platform**.
        3. **Fully NEW Design & Construct** (The materials from the original hull may but are NOT required to be used in any form as part of any reconstruction.)

    4. All expenses associated with hull redesign, modification, or reconstruction are considered part of the team's budget limitations.
    5. The team is required to designate a "**Primary" competition vessel** which must meet the following minimum design requirements:

        1. **Dimensional Envelope** - The vessel must not exceed the overall envelope dimensions of the originally provided hull. This is to be considered a "cubic" measurement of X-Y-Z:

            1. Length Overall (LOA) being the total length from bow (front) to stern (rear), must NOT exceed 144 inches or 12 feet

TrineOnline →
About Trine
News
Give
Athletics
Alumni
International
Academics
Become a Student
Life at Trine
Student Support
TRINE UNIVERSITY

2. Beam, which is the maximum width, must NOT exceed 40 inches
3. Overall Height, which is the combined total of Freeboard and Draft, must not exceed 16 inches. Draft is the measurement from the waterline to the keel (bottom) of the vessel.
4. The Freeboard, which is measured from the “deck” of the vessel to the water line, must be measured at 5-7 inches at the beginning of the competition. This is defined by the “orange” line on the originally provided hull.

6. **Payload Capacity** – The vessel must support a **minimum payload of 250 lbs**

TrineOnline →

About Trine
News
Give
Athletics
Alumni
International
Academics
Become a Student
Life at Trine
Student Support

1. The Competition vessel; either as provided, modified, or constructed, must hold a minimum of 250lbs. of payload. The payload is weight that is added in addition to any of the vessels control systems, propulsion systems, power challenge systems and other subsystems.
2. The weight is to be added to reach the 5-7 inch Freeboard as defined in the Dimensional Envelope requirements. At the beginning of the challenge the cessel will be reviewed when free from tether and contact to see if the vessel freely floats within the 5-7 inch freeboard line ("orange band")
3. After the vessel is underway the freeboard does NOT have to be maintained throughout the competition.

7. **Multi-Vessel Configurations**

1. Teams may choose to design and deploy **multiple unmanned surface vessels (USVs)** and/or unmanned aerial vehicles (UAVs).
2. **Envelope Limitations**: The combined footprint of all USVs, when deployed together, must not exceed an envelope of **3 times the size**

of the original LPV hull, as defined in the Dimension Envelope.

3. **Number of Vessels**: There is **no limit** to the number of USVs or UAVs a team may deploy, provided they comply with envelope, safety, and budgetary restrictions.
4. **Start Condition**: At the start of the competition, **all USVs and UAVs must be coupled, docked, or physically touching** the primary LPV. This configuration will be referred to as the **"marsupial vessel."**

8. **USV-UAV Deployment & Recovery**:

1. Once deployed, USVs may operate independently but must **all reconvene** before course completion.
2. Reconvening does **not** require the USVs to connected or to be touching but requires the multiple vessels to reorganize in the finishing position as a group.

9. **USV-UAV Operations & Safety**:

1. Once deployed, USVs may operate independently but must **all reconvene** before course completion.
2. Reconvening does **not** require the USVs to connected or to be touching but requires the multiple vessels to reorganize in the finishing position as a group.

TrineOnline →

About Trine
News
Give
Athletics
Alumni
International

Academics
Become a Student
Life at Trine
Student Support

TRINE UNIVERSITY

1. On a specified **competition target**, or
2. Returning to the **primary LPV**.
3. UAVs NOT landing on one of these designated locations WILL result in point deductions or disqualifications. The intent of UAV use in the competition is not attritable or kamikaze in nature

3. All USVs & UAVs must be turned on and calibrated while the LPV is safely tethered to the dock. Drones may have the gyro calibrated while on the dock and land, absolutely NO flights may be launched from the dock or from DNR property
4. The Relay Distance for any USVs & UAVs must be provided to the safety inspector and event staff to verify that they are an acceptable distance for the competition. The minimum relay distance from the communication position will be a distance able to cover the entire course.
5. USVs & UAVs will be reviewed and accessed for any potential unsafe conditions.
6. All teams will be required to notify the judges and event team of any USV or UAV modifications or additions made after the initial safety inspection. A safety assessment will be completed following any changes.

TrineOnline →

About Trine

News

Give

Athletics

Alumni

International

Academics

Become a Student

Life at Trine

Student Support

TRINE UNIVERSITY

10. **Primary Vessel Challenge Requirement**: Regardless of multi-vessel deployment, the team's designated **primary LPV** must complete both:
    1. Slalom Challenge (#1)
    2. Evade / LiDAR Challenge (#3)

8. **Budgetary Requirements**
    1. **Direct Expense Limit**: Direct expenses (including but not limited to hull modifications, sensors, hardware, and competition-related components) may not exceed **$5,000 in additional costs**.
    2. **Gift-in-Kind & University Resources**: Teams may use donated items and/or existing university resources, provided the **fair market value (FMV)** of these contributions does not exceed $5,000.

3. **Total Allowable Budget**: The maximum allowable project budget may NOT exceed **$10,000**, inclusive of:

    1. Direct purchases
    2. Gift-in-kind contributions
    3. University-provided resources (valued at FMV)

4. **Excluded Expenses**: The $10,000 limit does **not** apply to:

    1. Travel, food, or lodging
    2. Student labor or faculty time
    3. Gifted Services

5. **Budget Documentation**: Teams must submit a **budget report** to prove compliance to the budgetary requirements that includes:

    1. Itemized expenditures for direct purchases
    2. Documentation of gift-in-kind donations with FMV
    3. Documentation of university resources with FMV

6. **Compliance**: Non-compliance with these design or budgetary requirements may result in penalties, loss of points, or disqualification.

9. There is no maximum number of student participants per team criteria; student participants must comply to the member eligibility criteria.
10. Additional rules for the final event including location, arrival times, safety, rubrics, etc. will be provided prior to the AIMM ICC.

TrineOnline →

About Trine

News

Give

Athletics

Alumni

International

Academics

Become a Student

Life at Trine

Student Support

TRINE UNIVERSITY

Right to Revise Rules

Trine University and NSWC Crane reserve the right to revise the rules at any time by providing the participants notification of revisions in the form of bulletins, revised editions of the rules, or announcements at the competition.

Acceptance of Rules

All persons or groups selected to participate in the competition are expected to know and accept the Rules. Participation in the competition shall constitute acceptance of them.

TrineOnline →

About Trine

News

Give

Athletics

Alumni

International

Academics

Become a Student

Life at Trine

Student Support

Team Member Eligibility

Anyone who has attended the participating university as an undergraduate full-time student in one of the past two semesters, or two of the past four quarters, or part time for the past year, is eligible to serve as a team member. Graduate students can serve as advisors and mentors but should not be active members of the team, unless complying with the prior student criteria.

Supplies

For this competition, teams will use the supplied Low Profile Vessel and sensor kit. The teams can make small changes to the boat design, but all material should be easily sourced at local hardware stores, or a cost-effective online supply source.

TRINE UNIVERSITY

The teams will also be provided with a sensor suite. It is up to the individual teams to add on any extra sensors they deem necessary, and program their own algorithms to make the sensor suite work to the individual LPVs.

*Additional costs incurred must comply with Rule 8 outlined above.

Scoring

The competition will be judged in three Topic Areas (Technical Report, Presentation, and Final Competition). Awards for the Top Team will be awarded for each Topic Area. The combined efforts of all three Topic Areas will yield the standings for the overall winner.

Documents

Challenge Brief
Competition Course Details
Challenge Competition Scoresheet
Score Log
Technical Design Report
Design Presentation
Buoy Images
Vehicle Safety Inspection
Sensor Package
Supplied Project Materials

TrineOnline →

About Trine

News

Give

Athletics

Alumni

International

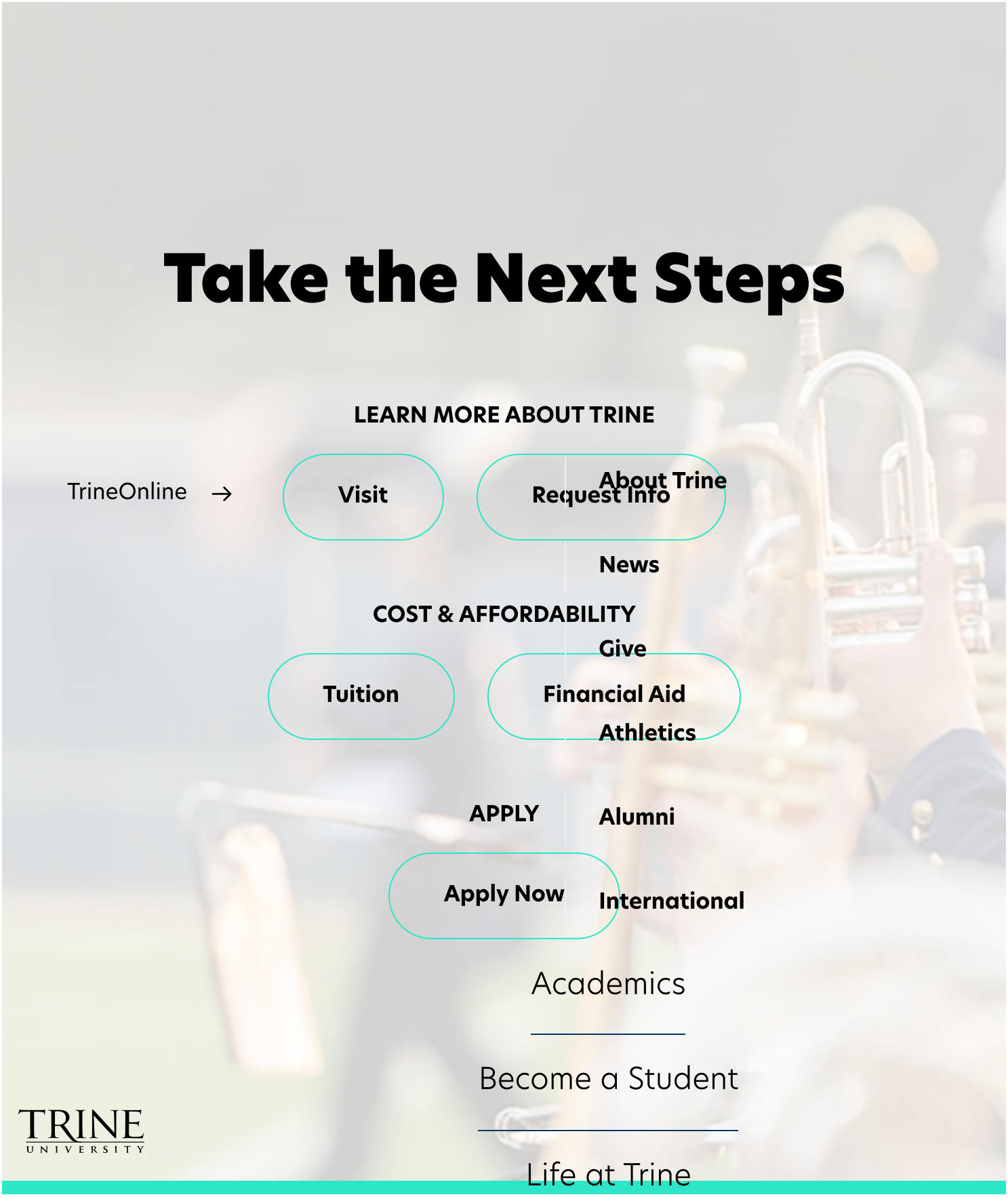
Academics

Become a Student

Life at Trine

Student Support

TRINE UNIVERSITY



Take the Next Steps

LEARN MORE ABOUT TRINE

TrineOnline →

Visit

Request Info

COST & AFFORDABILITY

Tuition

Financial Aid

APPLY

Apply Now

About Trine

News

Give

Athletics

Alumni

International

Academics

Become a Student

Life at Trine

Student Support

TRINE UNIVERSITY

TRINE UNIVERSITY

TrineOnline →
260.665.4100
800.347.4878

One University Avenue
Angola, IN 46703

About Trine

News

Give

Athletics

Alumni

International

HELPFUL LINKS

Employment

Contact Us

Maps & Directions

Academics

Become a Student

Life at Trine

Student Support

LOGINS

MyPortal

Moodle

Webmail

AFFILIATES

TrineOnline

Trine Innovation 1

Education One

Zollner Golf Course

T. Furth Center for Performing Arts

Trine Broadcasting Network

Copyright © 2026 Trine University. All Rights Reserved.

Privacy Policy

Title IX

Student Right to Know